

IoT-based AQI Estimation using Image Processing and Learning Methods

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Abstract—Air pollution is a concern to the health of all living beings. It is essential to check on the quality of air in the surroundings. This article presents an IoT-based real-time air quality index (AQI) estimation technique using images and weather sensors on Indian roads. A mixture of image features, i.e., traffic density, visibility, and sensor features, i.e., temperature and humidity, were used to predict the AQI. Object detection and localization-based Deep Learning (DL) method along with image processing techniques were used to extract image features while an Machine Learning (ML) model was trained on those features to estimate the AQI. In order to conduct this experiment, a dataset containing 5048 images along with co-located AQI values across different seasons was collected by driving on the roads of Hyderabad city in India. The experimental results report an overall accuracy of 82% for AQI prediction.

Index Terms—Air Quality, CNN, Edge computing, Machine Learning.

I. INTRODUCTION

Air pollution consists of chemicals and particles suspended in the air. Exposure to such air can cause very mild problems like irritation in the eyes and nose, ranging up to fatal diseases like cancer in the long term [1]. The sources of pollution can be stationary (chimneys, construction sites) and mobile, such as vehicles. Due to the increasing industrialization and urbanization, air quality remains an essential concern for all living beings' good health and hygiene.

Generally, there are only a few expensive but good quality air pollution monitoring stations in a city. Dense stationary deployment of internet-of-things (IoT)-based low-cost air pollution monitors can help in increasing the spatial resolution of the pollution data [2–4].

Electronic gadgets are also available in the market that can monitor air quality and report the concentration of particulate matter (PM) from which AQI can be computed. These devices, however, demand maintenance with time due to their limited lifetime. For example, SDS011 by Nova Fitness is one of the most widely used laser-based PM sensors. The datasheet [5] of this sensor claims a lifetime of 8000 hours which is roughly one year.

Instead of using a pollution monitoring, the AQI can also be determined with the help of image processing-based technique. Image based AQI calculation can serve the purpose in rural and sub-urban areas where sensors are unavailable. In general, sensors require regular maintenance. In addition, having an image-based AQI estimation allows portability through smartphones as a user can capture a traffic image and estimate the AQI

in real-time. In this paper, a method is proposed to estimate the AQI levels using traffic images and weather parameters instead of pollution monitoring sensors.

Related work: In recent years, some work has been done in the field of air quality measurements using DL paradigms. Few of them [6–8] use deep learning based algorithms on meteorological data-based air quality estimation, while some of the work incorporates images from different locations to compute air quality [9, 10]. [9] extracts various numerical and categorical from an image using image processing algorithms and applies Support Vector Regression (SVR) to train and predict PM values. Their dataset contains 6587 images collected from a fixed scene with respective PM_{2.5} value, weather data, and geographic location spanning three cities of China. The SVR model performs good for the two cities but fails on the third city because of the narrow range of PM_{2.5} values.

[10] uses CNN architecture along with transfer learning on a manually created dataset. The dataset contains a total of 591 images collected across different seasons from a Beijing tourist website with their respective PM_{2.5} values. The air quality is categorized into three levels, i.e., good, moderate, and severe. Overall, the proposed method achieved 68.74% classification accuracy. However, [10] did not justify achieving the mentioned accuracy and what precisely the CNN model learned from the images to predict the AQI.

The main contributions of this article are as follows:

- An IoT based novel methodology is proposed to estimate the real-time AQI into five levels using traffic images and weather parameters. To the best of authors' knowledge, this article is the first of its kind to achieve this on Indian roads.
- An entirely new traffic dataset is collected on Indian roads containing 5048 images and related weather data with co-located ground truth PM values. The dataset contains samples across the Indian city of Hyderabad in different seasons.
- The proposed method achieved overall 82% accuracy considering PM variation due to season. We show a significant improvement in the accuracy of AQI estimation using images when compared with existing work [10].

The structure of this paper is as follows: Section 2 briefly describes the hardware setup. Section 3 elaborates the details of measurement campaign and the resultant dataset. Section 4 presents the proposed methodology that was used to determine AQI using the dataset. Results are presented in Section 5 while

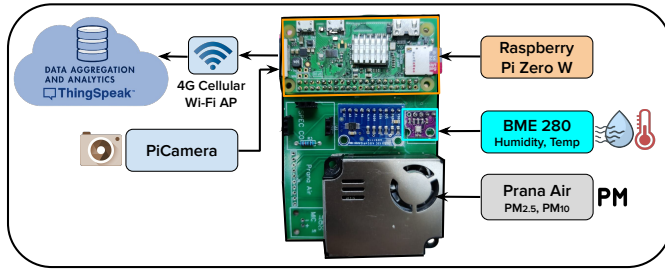


Fig. 1: Block and circuit diagram of hardware setup.

Section 6 concludes this paper.

II. HARDWARE SETUP

Fig. 1 shows the block architecture of the hardware developed for this experiment. A Raspberry Pi Zero W (Rpi0) microcontroller unit (MCU) and a PiCamera are connected to it to capture and process the vehicle images. The other sensors that were interfaced with the MCU include BME280 for temperature and humidity. A *Prana Air* [11] sensor was used for measuring the $PM_{2.5}$ and PM_{10} concentrations. It is a reliable PM sensor, as shown in the study done by [12]. The data collected from *Prana* sensor was used to calculate the AQI which also served as the ground truth for the ML algorithm developed for this experiment. The hardware setup is capable of sending the processed data into a remote server, making it suitable for edge computing. A sample from each sensor was collected once in every 30s by the MCU. The sample is processed using the methodology defined in the upcoming sections. Cellular 4G-based Wi-Fi access point was used to send the data to the remote server.

III. MEASUREMENT CAMPAIGN AND DATASET

With the help of the hardware setup mentioned in the previous section, a traffic dataset was collected, containing images of traffic and the measurement of pollution levels. The device was placed on top of a car. The car was driven

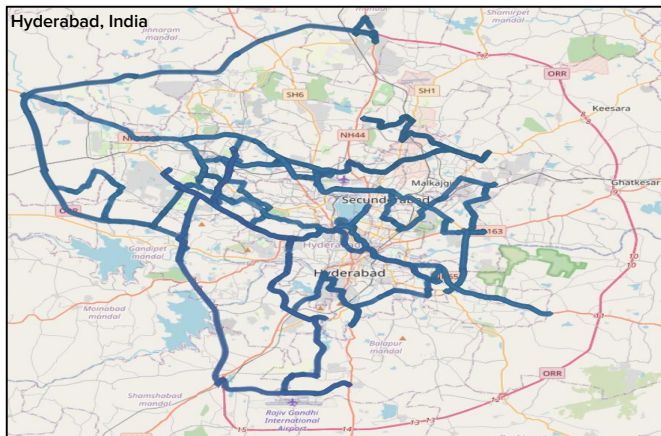


Fig. 2: Street view of routes traveled during measurement campaign (Total distance = 1000 km).

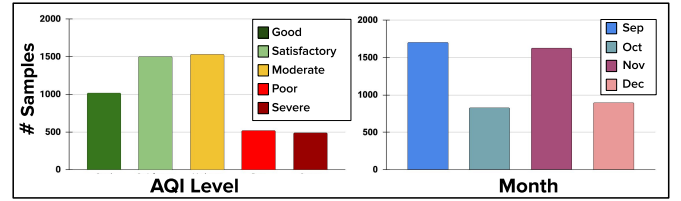


Fig. 3: Left: Frequency of the AQI levels in the collected dataset. Right: Frequency of samples collected across months. (Best viewed in color).

during the daytime and captured variations, including different scenarios (urban and sub-urban areas), traffic conditions, and pollution levels. The dataset was captured across Sep'21-Dec'21, comprising two seasons, monsoon and winter. The attempt was to get a diverse dataset. A total of 5048 samples were collected in this duration. Datapoints collected between Sep'21 and Oct'21 were considered for the monsoon season. The rest of the data collected during Nov'21- Dec'21 was accounted for the winter season. Fig.2 shows the routes traveled during this campaign in the metro city of Hyderabad, India. Each captured image is associated with co-located respective sensor values, i.e., temperature, humidity, $PM_{2.5}$, and PM_{10} measurement. The AQI level is computed using the $PM_{2.5}$ and PM_{10} values as per the Central Pollution Control Board, India [13], and categorized into five classes which are as follows: 1. **Good** (0 - 50) 2. **Satisfactory** (51-100) 3. **Moderate** (101-200) 4. **Poor** (201-300), and 5. **Severe** (>300). The distribution of the collected data in terms of the AQI level and month is shown in Fig. 3.

IV. PROPOSED METHODOLOGY

We propose a novel methodology to estimate AQI using traffic images for Indian roads. A pipeline for the same is shown in Fig. 4. Firstly, the image features are extracted using DL and image processing methods and concatenated with sensor features to generate a feature dataset. Further, an ML model is trained on the feature dataset to estimate the AQI level.

A. Image Features Extraction

The central idea of this work is to estimate the air pollution i.e. AQI based on the traffic images. Hence, features were derived from the collected images to use it further for AQI calculation. From the given image, all the pollution-emitting vehicles were detected and their respective count was used as an image feature. To detect the vehicles from a given image, You-Only-Look-Once version 5 (YOLOv5) [14] was trained on Indian Driving Dataset (IDD) [15].

IDD is object detection and localization dataset that includes images covering highways and lanes with various traffic scenarios and illumination conditions. The objects in the images are annotated (drawing a bounding box around the object) finely with their respective classes. Although the IDD has 34 unique labels containing every possible object in the traffic,

TABLE I: Distribution of type of vehicles in the detection and localization dataset.

Vehicle	Bike	Car	Truck	Bus	Rickshaw
Count	103608	90520	27837	18745	32280

our use case is limited to identifying objects which contribute to air pollution. Hence, five pollution-emitting vehicle classes were picked which are as follows: 1. **Motorcycle** 2. **Car** 3. **Truck** 4. **Bus** 5. **Autorickshaw**. As the classes are limited, all the images containing the above-defined classes from the IDD were selected. The rest of the images were discarded, resulting in a subset of the IDD. There were a total 37869 images in the resulting dataset for which the object frequencies are shown in Table I.

To quantify the vehicles present in the image, YOLOv5 algorithm was used. YOLOv5 is an object detection and localization algorithm which uses Convolutional Neural Network (CNN) as the feature extractor to detect and localize multiple objects in a given image. The final output of YOLOv5 for a single image is: detected objects (classification) and their bounding box (regression). YOLOv5 was trained on the above-defined custom IDD.

B. Image Visibility Score Calculation

As the camera can only take images of the traffic and road condition in front of the test-vehicle, the information received by the image is limited in terms of scope. It is essential to capture the air pollution generated by the other elements, e.g. constructions (road, buildings etc.), fire etc as well. To capture the essence of pollution caused by other sources, the visibility of the image is computed using Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE) [16] which is a no-reference Image Quality Assessment (IQA) metric. As visibility is a subjective matter, a human evaluated dataset named TID2008 [17] which has 1700 images and their respective quality scores is used as image visibility score reference. The output of BRISQUE algorithm for a given image is a number between 0 to 100, where 0 signifies the best, and 100 signifies the worst visibility. An example of image visibility metric calculation can be seen in Fig. 4.

C. Feature Vector Generation

A feature vector is generated corresponding to each sample defined in the dataset collected in Section III. The steps are as follows:

(a) **Image features:** Firstly, the image is passed to the trained YOLOv5 model, which detects the count of pollution-emitting vehicles present in the image. The detection includes the count for each type of vehicle, i.e., bus, car, truck, motorbike, and autorickshaw (total 5), and is treated as the image feature vector. After that, the visibility score of the image is computed using the BRISQUE algorithm. The output of this algorithm is a single number and contributes as an image feature vector.

(b) **Sensor features:** Finally, the corresponding sensor values, i.e., temperature and humidity, are concatenated to the

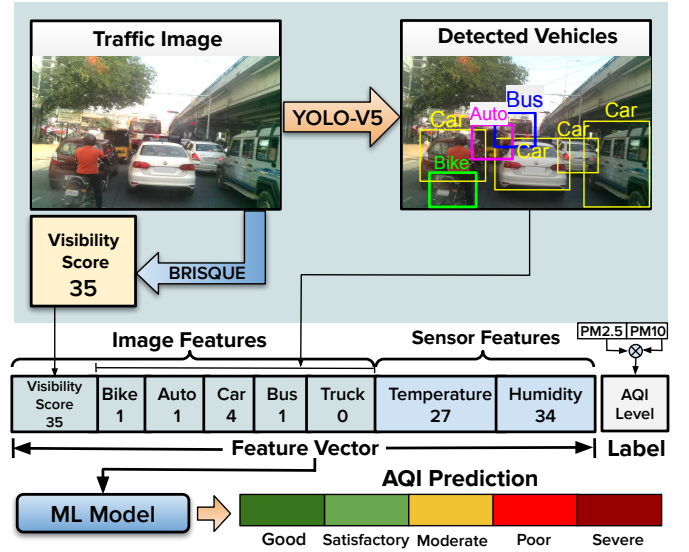


Fig. 4: Algorithmic pipeline of the proposed method. Firstly, the image is passed to the trained YOLOv5 and BRISQUE algorithm to generate the image features. Then it is concatenated with corresponding sensor features to generate a complete feature vector with the corresponding label. Finally, an ML model is used to train and detect the AQI into five categories. (Best viewed in color)

image feature vector to produce the complete feature vector of size 8×1 for the given sample. An example of the feature vector is shown in Fig. 4.

(c) **Label:** The associated $PM_{2.5}$ and PM_{10} concentration values obtained from the reference PM sensor are used to generate the label for a given sample. An AQI value is computed using these sensor detected values and categorized into one of the five AQI levels mentioned in section III. The categorized AQI level is used as the corresponding label.

After processing the above-defined steps for all the samples in the dataset, a $m \times 8$ sized data matrix M is obtained, where m is the total number of samples present in the dataset. A $m \times 1$ sized vector y containing the corresponding labels is also obtained.

D. Training

With the help of the dataset M , and the corresponding label vector y , an ML model was trained to classify the samples into five different AQI levels. As this is a supervised learning problem, where all the dataset features have continuous values, a classification-based ML model was trained to predict the AQI for a given sample. To choose the best performing ML model, three different ML models that best suit the data were experimented: 1. Random Forest (RF) 2. Support Vector machine (SVM) and 3. Multi Layer Perceptron (MLP). Each of the models has five classes (AQI levels) as output.

Firstly, standard normalization was performed on all the dataset features as part of feature engineering. This preprocessing step ensures that the feature values are centered around the mean with unit standard deviation. After this, the normalized dataset was split into the training and validation part using K-

fold cross-validation strategy with 10 folds. This step ensures better model generalization ability than standard train/test split and minimizes the occurrence of bias estimation. The ML model was trained and validated on each fold separately. To evaluate the ML model, the average of the metrics (accuracy and F1 score) of the validation part across all the ten folds were calculated.

Detection: To compute the AQI for a given traffic location, the PiCamera captured the image, which is passed to the YOLOv5 and BRISQUE algorithm to compute the count of each type of vehicle and image visibility score respectively. The sensors provided the corresponding temperature and humidity values. After this, using the image and sensor features, a feature vector of size 8×1 was created and passed to the trained ML model to detect the AQI level. All the computation related to AQI estimation (YOLOv5, BRISQUE and ML model) are performed on Rpi0 itself and only the estimated AQI is sent to the remote server, thus making it suitable for edge-computation.

V. EXPERIMENTS & RESULTS

The YOLOv5 model was trained for 25 epochs and converted into a *TensorFlow Lite* [18] model to run on Rpi0. Fig. 5 shows evaluated metrics of the trained YOLOv5 model. An example of YOLOv5 output can be seen in Fig. 4.

For RF model, the number of decision trees was set to 100, and the split criterion was *entropy*. Tree pruning mechanisms were used to avoid overfitting. For SVM model, the regularization parameter was set to 5, and the *RBF* kernel was used. In the case of MLP model, three hidden layers, each having 20 neurons with *ReLU* activation function, was used to train for 100 epochs. All the ML models were implemented using the *Scikit-learn* [19] ML library. To detect one sample using the methodology proposed, the Rpi0 took around 20 seconds.

Table II shows the R^2 score of the features with respect to the AQI. It can be seen that all features have a positive

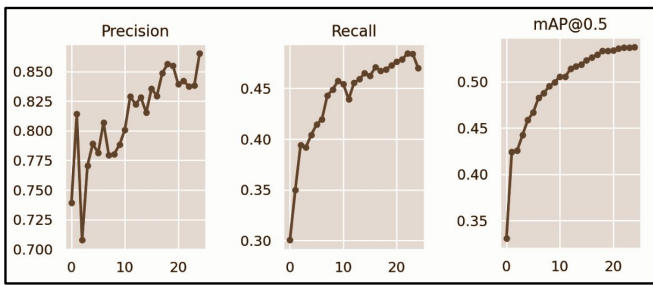


Fig. 5: YOLOv5 training results. X-axis of each graph represents number of epochs and Y-axis represents respective values. Final results at epoch 25 report 86.54% precision, 47% recall and 53.75% $mAP@0.5$.

TABLE II: R^2 score of features w.r.t AQI

	Vehicles	Temperature	Humidity	Visibility
AQI	0.33	0.29	0.31	0.41

TABLE III: Performance of various methods on overall and season-wise data. Note that the ML methods (SVM, MLP, RF) were trained on features extracted using method described in section IV-C.

Method	Monsoon		Winter		Overall	
	Acc	F1	Acc	F1	Acc	F1
SVM	0.86	0.85	0.74	0.72	0.77	0.76
MLP	0.90	0.89	0.78	0.74	0.79	0.78
RF	0.91	0.90	0.80	0.78	0.82	0.81
CNN [10]	0.71	0.70	0.61	0.61	0.67	0.65

correlation with the AQI in the range of 0.2-0.4. This indicates the partial dependence of the AQI on these features.

The ML models were trained and validated for the dataset mentioned in Section III. As there are seasonal variations in PM values [20], we trained three different models each for: 1. Monsoon dataset (samples collected between Sep'21 - Oct'21) 2. Winter dataset (samples collected between Nov'21 - Dec'21), and 3. Overall dataset (combining Monsoon and Winter dataset). The results obtained for all three datasets are presented in Table III.

For the overall dataset, the RF model achieved an accuracy of 82% and an F1-score of 81%. For the data points of monsoon season, it is observed that the RF classifier performs the best with an accuracy of 90.32%. The main reason for this relatively high accuracy is better training of the model as there are significantly high number of data points with low AQI values in monsoon season. Hence, most of the data points belong to the first two categories. This is more evident from the Fig. 3 that shows the category-wise distribution of dataset. Categories named as "Good" and "Moderate" account for 50% of the collected data points, which leaves significantly less room for the misclassification of the samples. On the other hand, RF is the best performing model for Winter dataset as well, with an accuracy of 80.14%. It is relatively low as compared to the monsoon season as the data is spread over all categories of AQI, which also increases the chance of misclassification.

To compare our proposed method with the existing work, we applied the method proposed in [10], which used plain CNN on each of the three datasets mentioned above and observed an overall accuracy improvement of 15%. The main reason behind this improvement is how features from images are extracted. As a traffic image can have different objects, our work emphasizes focusing only on those responsible for air pollution. On the other hand, applying plain CNN on the images straight-forward fails to identify this paradigm.

VI. CONCLUSION

This article introduced a simple yet efficient image-based AQI classification technique on an IoT device using a mixture of ML and DL-based supervised learning algorithm. Experimental results show accuracy up to 90% for the AQI classification. Additionally, a feature-rich dataset was created to be released in the public domain to promote further research. Currently, the work is limited to predicting the AQI only under daylight setting, and the authors plan to scale it to nighttime

as well as to collect data for the summer season. Authors also plan to collect data and predict AQI for different cities as a future work.

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